

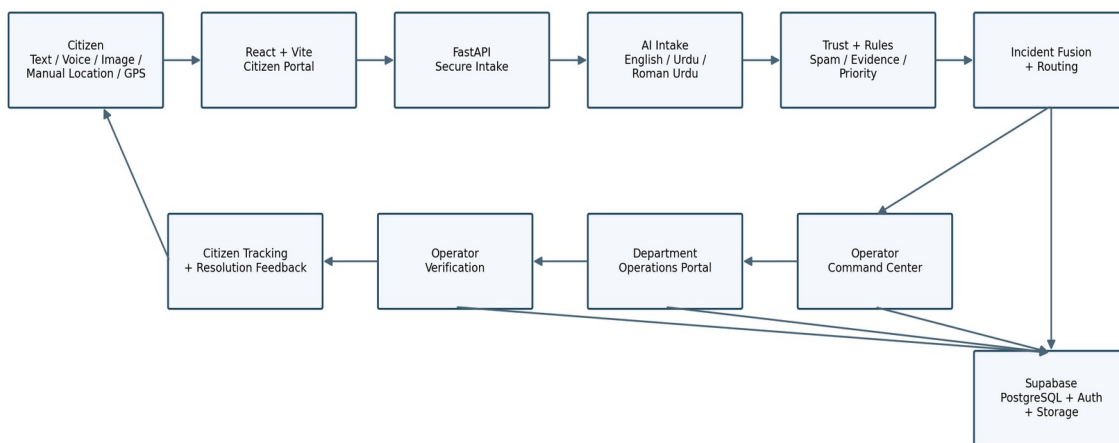
CivicOps AI

Product Requirements Document (PRD) - Final Hackathon Submission

Inclusive Community Incident Intelligence & Response Coordination Platform

Final Hackathon Submission | October 2026

CivicOps AI - High-Level System Architecture



Prepared by Team CivicOps AI

Judge's Quick View

What a judge should understand within the first minute

Core Problem	Core Innovation	Human Control	Impact
Fragmented, multilingual civic complaints become duplicate or incomplete tickets.	Separate report records from incidents, then fuse related reports into one operational issue.	AI structures and recommends; operator verification governs assignment and final closure.	Clearer prioritization, department accountability, traceable progress and citizen feedback.

Key distinction: CivicOps AI is not only a chatbot or complaint form. It is an end-to-end incident coordination workflow.

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1. Executive Summary

CivicOps AI is a multilingual civic incident intelligence and response coordination platform designed to convert fragmented citizen complaints into structured, traceable, human-supervised operational incidents. It is designed for environments where citizens may use English, Urdu, or Roman Urdu, may have limited digital literacy, and may not always be able or willing to share GPS coordinates.

Unlike a basic complaint portal, CivicOps AI separates individual reports from real-world incidents. Multiple citizen submissions can be linked to one incident, while evidence confidence, spam risk, priority, routing, department action, operator verification, and citizen resolution feedback are tracked as distinct parts of the workflow.

- Citizen reporting through text with optional GPS and manual location/landmark entry.
- Multilingual AI intake for English, Urdu, and Roman Urdu.
- Conservative incident fusion so duplicate reports do not become duplicate operational incidents.
- Rules and trust signals for priority, evidence confidence, and spam risk.
- Human-controlled operator decisions for verification, assignment, response-plan approval, and final resolution.
- Department portal for assignment acceptance, work updates, and completion submission.
- Citizen tracking and final resolution feedback after operator-confirmed closure.

2. Problem Statement

Civic problems such as sewerage overflow, water-supply failures, road damage, waste accumulation, street-lighting faults, utility issues, and local public-safety concerns are often reported through disconnected channels. The same incident may be reported by multiple people using different wording, while authorities may receive incomplete location details, duplicate complaints, exaggerated urgency, or malicious/spam submissions.

Key challenges addressed by CivicOps AI:

- Fragmented and unstructured reporting.
- English, Urdu, and Roman Urdu language variation.
- Duplicate reports that may represent one incident.
- Difficulty distinguishing claimed urgency from operational priority.
- Weak trust signals and susceptibility to spam or manipulation.
- Lack of clear department handoff and department-to-operator feedback.
- Poor visibility for citizens after reporting.
- Dependence on GPS in many digital systems, despite users often preferring landmark-based location descriptions.

3. Product Vision and Objectives

3.1 Vision

Provide a practical civic-operations layer that allows ordinary citizens to report problems in natural language while helping authorized operators convert those reports into structured incidents that can be assigned, acted on, verified, and transparently closed.

3.2 Objectives

- Reduce duplicate operational tickets by fusing related reports into shared incidents.

- Make civic reporting usable without requiring technical language or GPS permission.
- Provide structured, explainable operational signals instead of relying on free-form AI judgment.
- Protect sensitive actions with human approval and role-based authorization.
- Create a complete operational loop from citizen report to department action to verified resolution.
- Keep the system understandable and demonstrable within a hackathon environment while using production-minded patterns.

4. Scope and Non-Goals

In Scope	Out of Scope / Future Scope
Citizen web reporting, tracking, multilingual AI intake	Direct government dispatch integration
Manual landmark text and optional GPS	Guaranteed offline synchronization
Incident fusion and report-to-incident linking	Native mobile application
Operator dashboard and human verification	Advanced predictive analytics
Department dashboard and work updates	Multi-department membership per user
Supabase persistence, auth and audit history	Production SLA / certification
MapTiler + Leaflet incident maps	WhatsApp/SMS integration unless added later
Citizen resolution feedback	Department completion-photo workflow if not yet enabled

5. Target Users and Roles

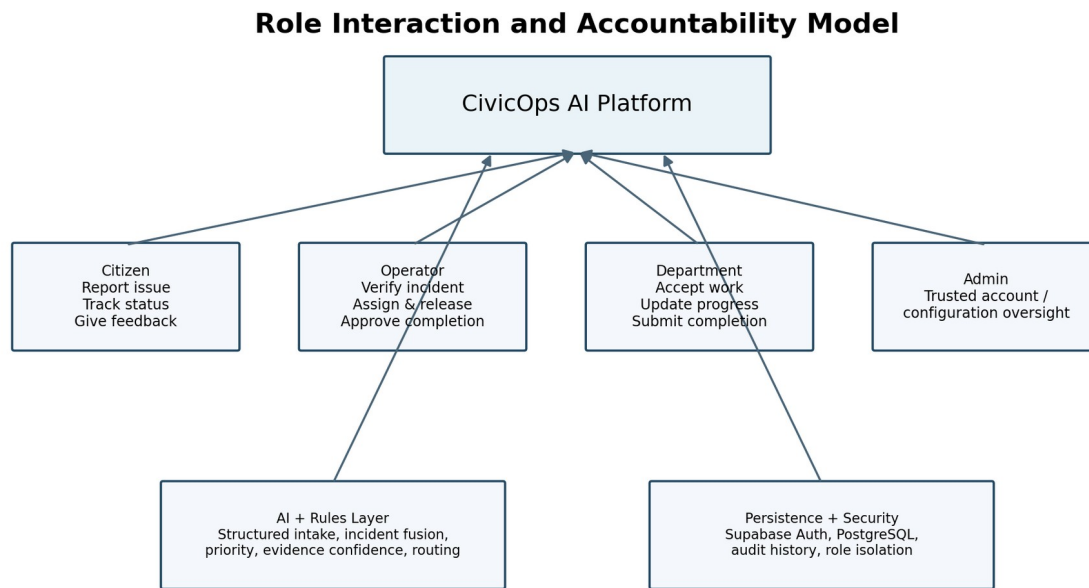


Figure 1. Role interaction and accountability model.

Role	Primary Responsibilities	Restricted From
Citizen	Submit reports, track own reports, provide eligible resolution feedback	Operational actions, department actions, admin controls
Operator	Verify incidents, approve response plans, assign & release departments, review completion, resolve/reopen	Public self-promotion to elevated role
Department	View own assigned queue, accept work, start work, add updates, submit completion	Other departments' queues, priority changes, final resolution, operator/admin controls
Admin	Trusted administrative oversight based on current implementation	Public self-assignment through signup

6. End-to-End User Journeys

6.1 Citizen Reporting Journey

- Citizen enters a description in English, Urdu, or Roman Urdu.
- Citizen may type an area/street/landmark without GPS.
- Citizen may optionally use current GPS location; manual landmark text is preserved.
- Report is validated, processed, persisted, and linked to a new or existing incident.
- Citizen receives a public report identifier and can track status.

6.2 Operator-to-Department Journey

- Operator reviews an incident and verifies it.

- Operator assigns and releases the incident to the selected department.
- Department sees the incident in its authorized queue and map.
- Department accepts the assignment, starts work, adds updates, and submits completion.
- Operator reviews department completion and either approves final resolution or reopens the incident.
- Citizen feedback is enabled only after final operator-confirmed resolution.

7. Functional Requirements

ID	Requirement	Priority
FR-01	Citizen can submit a report with text and optional location/media inputs.	Must
FR-02	Manual location/landmark must persist independently of GPS coordinates.	Must
FR-03	System supports English, Urdu, and Roman Urdu intake.	Must
FR-04	System returns structured category, summary, and reported urgency from AI intake.	Must
FR-05	Final operational priority is controlled by rules and system context rather than raw citizen urgency alone.	Must
FR-06	Related reports can link to one incident while original reports remain preserved.	Must
FR-07	System stores evidence confidence and spam risk as separate concepts.	Must
FR-08	Operator can verify, assign/release, and review incidents.	Must
FR-09	Department users can only access incidents assigned to their own department.	Must
FR-10	Department can accept, start work, add updates, and submit completion.	Must
FR-11	Only operator can confirm final RESOLVED state.	Must
FR-12	Citizen feedback is available only after final resolution.	Must

ID	Requirement	Priority
FR-13	Operator and department dashboards show incident locations on shared map when coordinates exist.	Must
FR-14	Incidents without coordinates remain usable in list/detail views.	Must
FR-15	Session handling must not repeatedly block/flicker on browser tab changes.	Must

8. Incident Intelligence and Decision Logic

8.1 AI Intake

The AI intake layer converts untrusted citizen text into a validated structured representation. AI is used for extraction and interpretation, not as an unrestricted authority. Structured outputs are validated with Pydantic models before use.

8.2 Reported Urgency vs Priority

Reported urgency represents what the citizen appears to claim. Priority is the operational severity used by CivicOps AI. The two are intentionally separate so a citizen cannot make an incident critical simply by writing 'urgent' or attempting prompt manipulation.

8.3 Incident Fusion

- Category compatibility.
- Text/lexical similarity.
- Geographic distance when coordinates are available.
- Specific landmark consistency when coordinates are absent.
- Time window and current operational state.
- Conservative behavior to avoid merging unrelated incidents.

8.4 Trust Signals

- Evidence confidence estimates how strongly the available evidence supports the incident.
- Spam risk estimates suspicious submission patterns and remains independent from evidence confidence.
- Idempotency prevents network retries from creating repeated report records.
- Human approval remains required for sensitive operational decisions.

9. Department Operations Workflow

Incident Lifecycle and Human Approval Gate

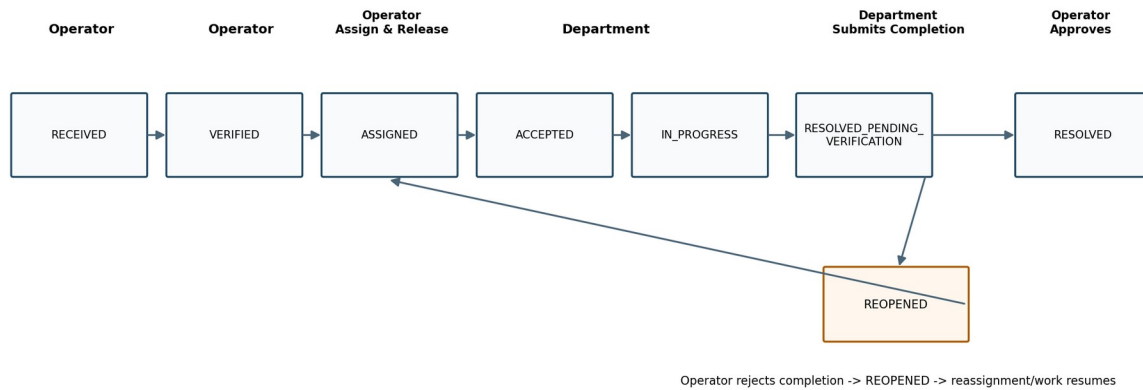


Figure 2. Incident lifecycle with department action and operator verification gate.

The department workflow closes the operational gap between assignment and real action. Department users are explicitly authorized against a trusted department membership and cannot inherit operator privileges.

State	Primary Actor	Purpose
RECEIVED	System / Operator	Incident exists and awaits review.
VERIFIED	Operator	Operator confirms incident is suitable for operational handling.
ASSIGNED	Operator	Assign & Release creates active department assignment.
ACCEPTED	Department	Department acknowledges responsibility.
IN_PROGRESS	Department	Department has started operational work.
RESOLVED_PENDING_VERIFICATION	Department	Department submits completion for operator review.
RESOLVED	Operator	Operator confirms final closure.
REOPENED	Operator	Completion rejected or incident requires further work.

10. Security, Trust and Abuse Protection

- Supabase Auth for identity and session management.
- Trusted profile role lookup on the backend.
- Explicit allowlists for CITIZEN, OPERATOR, ADMIN, and DEPARTMENT roles.
- Server-side department isolation; department users cannot view another department's queue.
- Pydantic request and AI-output validation.
- Idempotent report submission.

- Rate limiting where configured for sensitive/public endpoints.
- Prompt-injection-aware AI intake and untrusted-input treatment.
- Private evidence authorization.
- Atomic PostgreSQL RPCs for operational writes, status history, and audit events.
- Stale-update protection using expected version/timestamp checks.
- Final resolution requires human operator confirmation.

CivicOps AI is a hackathon implementation and should not be presented as a certified or fully production-hardened government system. The design demonstrates security-conscious patterns and human accountability.

11. Data and Persistence Model

Entity	Purpose
profiles	Trusted application role and department context.
departments	Configured operational departments.
reports	Original citizen submissions and structured intake results.
report_media	Private evidence metadata when media is provided.
incidents	Operational real-world incident records.
incident_reports	Links one or more reports to an incident.
incident_assignments	Department assignment history.
incident_status_history	Status transition history.
resolution_feedback	Citizen feedback after final resolution.
audit_logs	Traceability for important system and human actions.

Core relationship: reports remain independent source records, while incident_reports links them to incidents. This enables many reports to describe one operational incident without losing the original citizen submissions.

12. System Architecture and Technology Stack

CivicOps AI - High-Level System Architecture

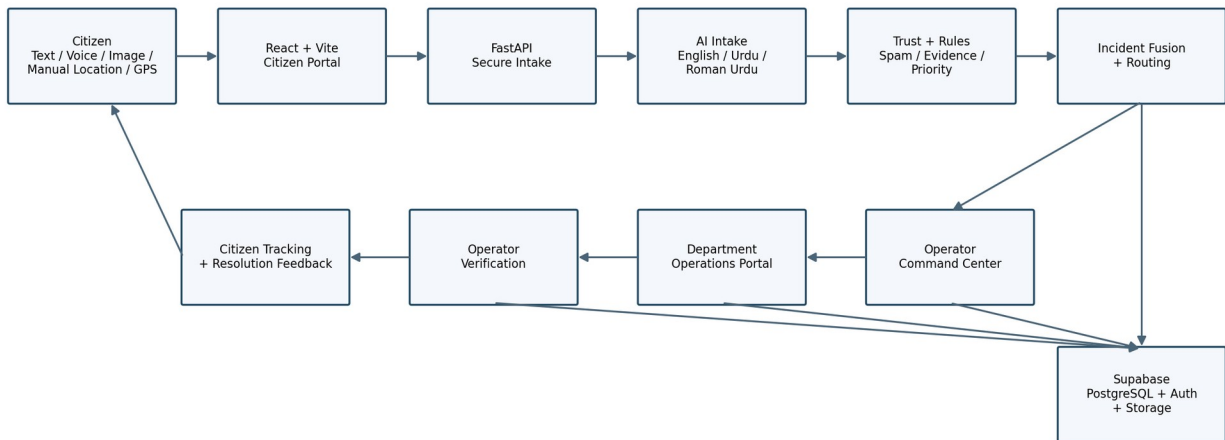


Figure 3. High-level CivicOps AI system architecture.

Layer	Technology
Frontend	React, TypeScript, Vite, Tailwind CSS
Maps	Leaflet / React Leaflet with MapTiler Streets raster tiles and OpenStreetMap attribution
Backend	Python, FastAPI, Pydantic
AI	Groq-hosted LLM structured intake
Database	Supabase PostgreSQL
Authentication	Supabase Auth
Storage	Supabase Storage where evidence storage is used
Deployment	Vercel frontend, Railway backend

13. UX, Accessibility and Pakistan-First Design

- English, Urdu, and Roman Urdu reporting support.
- Simple citizen-facing forms and status wording.
- Manual area/street/landmark entry is enough; GPS is optional.
- Maps support operational users but are not required for text-only incident handling.
- Responsive citizen, operator, and department views.
- Clear empty/error/loading states.
- Citizen feedback uses simple YES / PARTIALLY / NO choices.
- The design avoids requiring citizens to understand department structures or technical incident taxonomy.

14. API and Integration Requirements

Area	Representative API Capability
Reports	Create report, retrieve report, public-safe tracking
Authentication	Trusted current user context
Incidents	List/detail, allowed actions, status changes
Operator	Assign & Release, response-plan review, notes, final resolution review
Department	Own queue, detail, allowed actions, work updates, status progression
Dashboard	Summary and map-ready incident data
Media	Authorized evidence access
Feedback	Citizen resolution feedback

API contracts are typed and validated. Elevated actions are authorized server-side and must not rely on frontend visibility alone.

15. Non-Functional Requirements

Category	Requirement
Security	No public role elevation; secrets stay server-side; restricted evidence access.
Reliability	Operational writes should be atomic and preserve history/audit records.
Performance	Dashboard interactions should remain responsive; maps should not remount unnecessarily.
Usability	Citizen reporting must work without GPS and on mobile screens.
Maintainability	Shared map and API components should be reused across operator/department views.
Observability	Operational changes should be traceable through history and audit logs.
Graceful failure	AI/map/external failures must not corrupt report persistence or expose secrets.

16. Testing and Acceptance Criteria

- Citizen can submit manual-location-only, GPS-only, combined, and no-location cases where allowed.
- Session does not repeatedly block on tab focus or silent token refresh.
- Operator and department maps load through MapTiler and preserve queue-to-marker synchronization.

- Department A cannot access Department B incidents.
- Citizen cannot call operator or department mutation endpoints.
- Department cannot directly mark final RESOLVED.
- Assign & Release creates a valid department queue item.
- Department acceptance/work/completion survives refresh and persistence.
- Operator can approve completion or reopen incident.
- Citizen feedback remains blocked until final RESOLVED.
- Incident fusion handles coordinates and landmark-only reports safely.
- Frontend build, lint, backend tests, and regression tests pass before submission.

17. Deployment and Operations

Component	Deployment / Configuration
Frontend	Vercel
Backend	Railway
Database/Auth/Storage	Supabase
Map tiles	MapTiler browser API key restricted by allowed HTTP origins

17.1 Important Environment Variables

Frontend examples (exact names must match repository configuration):

- VITE_API_URL
- VITE_SUPABASE_URL
- VITE_SUPABASE_PUBLISHABLE_KEY
- VITE_MAPTILER_API_KEY

Backend examples include Supabase server credentials and GROQ_API_KEY; real secrets must never be committed to the repository.

Vite variables are embedded at build time, so frontend environment-variable changes require a redeploy.

18. Limitations and Future Scope

18.1 Current Limitations

- The system is a hackathon prototype, not a live government deployment.
- Operational departments and workflows are configured for demonstration and would require organization-specific onboarding.
- Reports without coordinates cannot appear as map markers, though their manual location remains available in text views.
- Advanced notification infrastructure is not part of the core implementation unless separately added.

18.2 Future Scope

- WhatsApp and SMS reporting/status notifications.
- True low-connectivity offline queue and later synchronization.
- Department completion-photo/evidence workflow.

- Multi-department membership and field-team assignment.
- Advanced incident analytics, hotspot trends, and service-level reporting.
- Government-system integrations and production-grade observability.
- Broader accessibility support and additional regional languages.

19. Team Structure

Order	Team Member	Responsibility
1	Muhammad Furqan Arshad	Team Lead, System Architecture & Integration
2	Ghulam Mohy Ud Din	Backend & AI Systems Lead
3	Khadeeja Ameen	AI Evaluation & Community Experience Lead
4	Muhammad Huzaifa Shamas	Frontend & Command Center Lead
5	Aleeba Pervaiz	QA & System Validation Lead
6	Muhammad Ali	Incident Intelligence & Operations Coordinator

20. Submission Summary

CivicOps AI demonstrates how Generative AI, deterministic operational logic, secure role-based workflows, and structured data can be combined into a practical civic-response platform. Its core value is not a chatbot interface; it is the complete transition from unstructured citizen input to a verified, assigned, department-handled, operator-confirmed, and citizen-reviewed incident lifecycle.

Citizen Report -> Structured Intelligence -> Incident Fusion -> Human Review -> Department Action -> Verified Resolution -> Citizen Feedback